



Enterprise AI Knowledge & Retrieval Platform

A private generative-AI platform for a large diversified holding group — a conversational assistant that answers questions over the organization's entire document corpus with citations, and a board-meeting copilot that prepares, fact-checks, and stress-tests decisions live. Powered by agentic Retrieval-Augmented Generation, a knowledge graph, and a fleet of specialized LLM agents.

DOMAIN

CORE

MODEL

PLATFORM

Enterprise Generative AI

Agentic RAG + Knowledge Graph

Private, permissioned SaaS

Azure · FastAPI · React

Confidentiality. Sanitized portfolio case study. The client identity, product names, brand assets, proprietary prompts, business rules, model deployment names, and internal source are withheld under NDA. Visuals are original architecture diagrams; no client screenshots, data, or credentials are reproduced. Technique parameters (fusion constants, embedding dimensions, token budgets) are generic; commercial figures are intentionally omitted.

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AT A GLANCE

CHALLENGE

A holding group's institutional knowledge was trapped in hundreds of documents. Keyword search returned files, not answers; generic chatbots hallucinate; board meetings ran on human memory; and every answer had to be grounded, cited, and permissioned.

SOLUTION

A private agentic-RAG platform: hybrid retrieval + rank fusion + two-stage reranking over an authorized corpus, grounded generation with citations, a two-phase-LLM knowledge graph, and a board-meeting copilot with an adversarial decision panel.

RESULT

Plain-English questions answered with sourced, permission-safe answers; meetings prepared, fact-checked, and stress-tested by AI; and a fully observable, measurable AI system delivered on Azure.

Hybrid + RRF

Keyword + dense vector + HyDE, fused ($k = 60$)

Grounded + cited

Answer-only-from-authorized-context

Knowledge graph

Two-phase LLM entity + relation extraction

Meeting copilot

Advocate / critic / arbiter / risk panel

CHALLENGE

Knowledge trapped, trust non-negotiable

A large, diversified holding group sits on an enormous, ever-growing body of institutional knowledge — board packs, financials, policies, legal agreements, rate cards, minutes, and market news — across hundreds of documents and dozens of subsidiaries. Several problems compounded:

- **Answers trapped in documents.** No way to ask a plain-English question and get a trustworthy, *sourced* answer. Keyword search returned files; generic chatbots hallucinate — unacceptable when a wrong number reaches a board.
- **Meetings ran on memory.** Preparing meant reading everything by hand; no way to fact-check a claim, surface a related past decision, or pressure-test a recommendation in the room; capturing outcomes was manual.
- **Knowledge was unconnected.** The relationships that matter — who supplies whom, which risk touches which portfolio, who owns which initiative — lived only in people's heads.
- **Trust, security & governance.** Every answer had to be grounded in *authorized* documents only, per user and per document, with nothing leaking across permission boundaries — and every response traceable and measurable.

SOLUTION

Two products over one agentic-RAG core

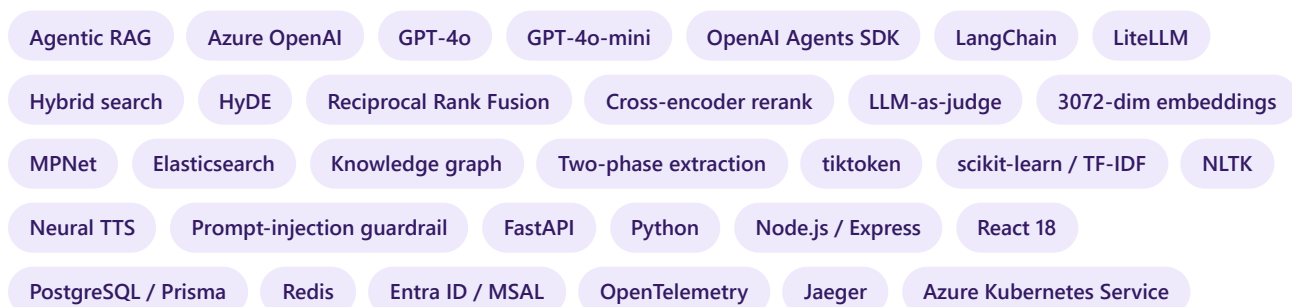
A conversational **knowledge assistant** (ask anything, get a grounded, cited answer from authorized documents) and a **board-meeting copilot** (prepare, fact-check, and stress-test decisions), both on one retrieval engine engineered for accuracy and trust — not a thin wrapper around a chat model.

- **Query understanding first.** One unified LLM agent classifies domain, extracts entities, expands keywords, and writes a HyDE hypothetical answer in a single call — consolidating five prior LLM calls into one.
- **Hybrid retrieval + fusion.** Boosted keyword, dense-vector kNN, and HyDE searches run in parallel and merge via Reciprocal Rank Fusion (k=60) — exact-match precision, semantic recall, and vocabulary-gap bridging together.
- **Signal-aware ranking + rerank.** Function-score (recency decay, doc-type priority, entity/domain boosts), then a cross-encoder and an LLM-as-judge contextual reranker refine the final context.
- **Grounded generation.** A document-level authorization prefilter is pushed into the search; an answer-only-from-context mode plus mandatory numbered citations mean the platform answers only from what a user may see, and shows its work.
- **Knowledge graph.** Two-phase LLM extraction (entities, then relationships constrained to them) builds a de-duplicated, synonym-merged graph that both visualizes connected knowledge and sharpens retrieval.
- **Meeting copilot.** Pre-meeting insight chains, live fact-checking and question generation, an advocate/critic/arbitrator/risk decision panel, and post-meeting action items, summaries, and an LLM-scripted TTS podcast.

Retrieval pipeline at a glance

```
Question + history → Unified query-understanding agent (domain · entities · keywords · HyDE)
Next-action router → retrieve / clarify / summarize / list      DB router → session vs historical
Hybrid retrieval (parallel): keyword (boosted BM25) + dense-vector kNN (3072-dim) + HyDE vector
    → Reciprocal Rank Fusion (k=60) → function-score (recency decay · doc-type · entity/domain boosts)
    → Rerank: cross-encoder → LLM-as-judge (relevance / fit / quality)
Prompt assembly: doc-level auth prefilter · citations · token-budget packing → Grounded GPT-4o answer
```

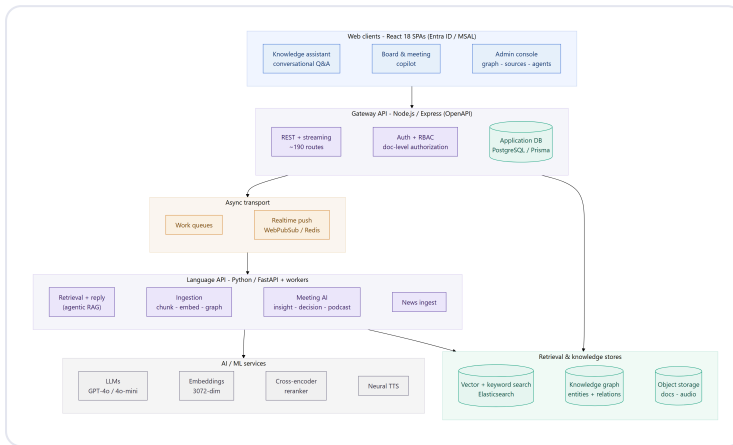
Technology



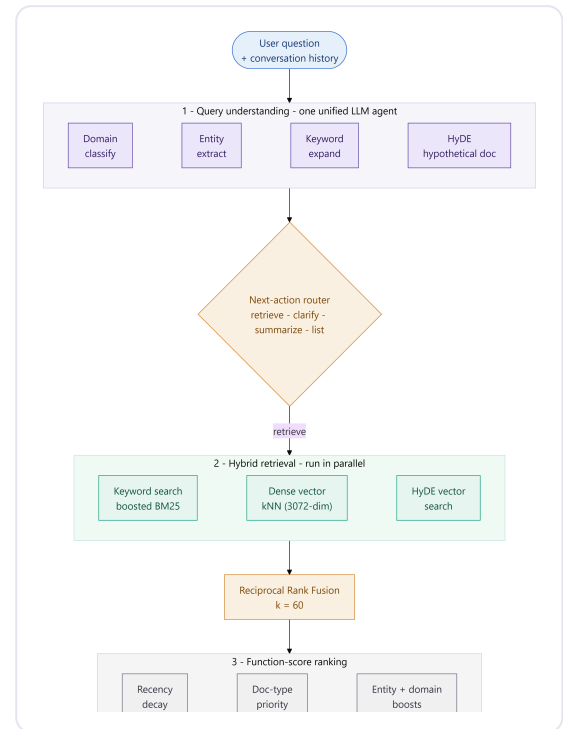
THE PLATFORM

System & the agentic-RAG pipeline

Original architecture diagrams — no client screenshots or data reproduced.

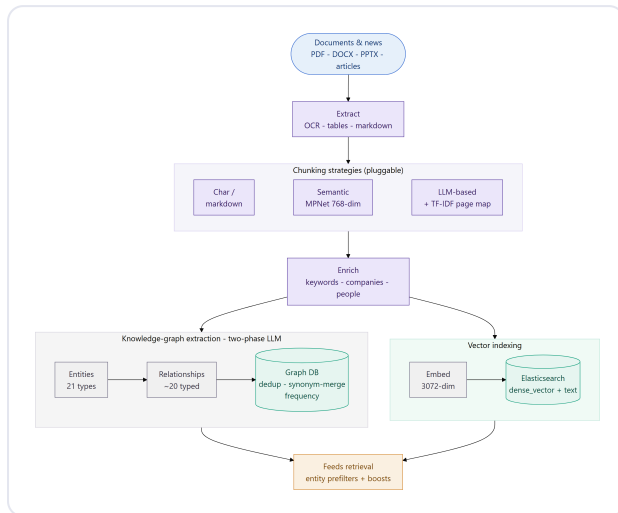


System architecture — React clients over a Node.js gateway that brokers to a Python/FastAPI language service and workers via async queues; AI/ML services and retrieval/knowledge stores sit behind the language service.

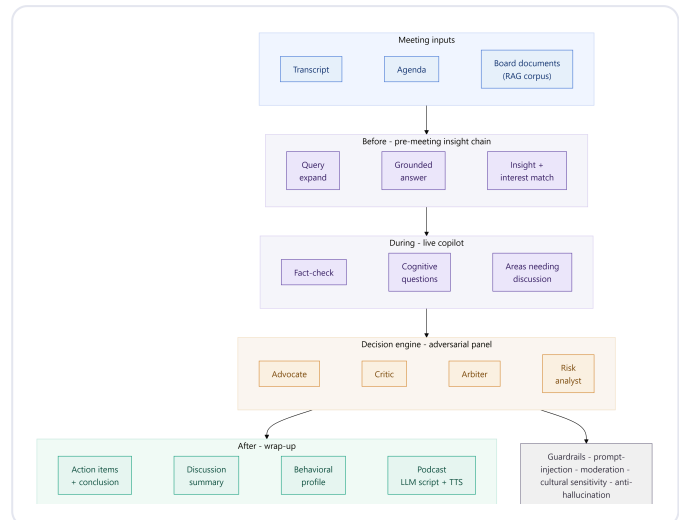


Agentic RAG pipeline — unified query understanding, action routing, hybrid retrieval, rank fusion, function-score ranking, and two-stage reranking before grounded, cited generation.

Knowledge graph & the meeting copilot



Ingestion & knowledge graph — pluggable chunking into both a vector index and a two-phase LLM graph extraction; the graph loops back to make retrieval smarter via entity prefilters and boosts.



Board & meeting copilot — pre-meeting insight chains, live fact-check and questions, an adversarial advocate/critic/arbiter/risk decision panel, and post-meeting wrap-up, all guardrailed.

RESULT

Trustworthy AI over the whole corpus

Sourced answers

Grounded, cited, permission-safe responses

AI in the room

Meetings prepared, fact-checked & stress-tested

Observable

Tracing + per-response eval metrics

- **Agentic RAG, not a chatbot wrapper.** Query understanding, routing, hybrid retrieval, fusion, ranking, two-stage reranking, budgeted assembly, and grounded generation are each engineered stages — accuracy designed in, not hoped for.
- **Grounded & permissioned by construction.** A document-level authorization prefilter pushed into the search, plus answer-only-from-context and mandatory citations, means answers come only from authorized sources — and always show their work.
- **A graph that improves retrieval.** Two-phase LLM extraction with cross-document dedup and synonym merging builds a living knowledge graph that both visualizes connections and sharpens search.
- **Cost-and-latency engineering.** Five query-understanding calls collapsed into one; cheaper models for routing and the flagship reserved for reasoning; parallel retrieval; token-budget packing that preserves fusion order.
- **A meeting copilot with an adversarial panel.** Advocate, critic, arbiter, and risk agents argue a recommendation before it reaches the room, alongside live fact-checking and an LLM-scripted TTS recap.
- **Observable, measurable, scalable.** OpenTelemetry tracing across gateway and workers, per-response token-and-entity metrics, offline evaluation tooling, and independently scaling queue-driven workers on Kubernetes.

Note on scope. Capabilities and technique parameters reflect the delivered system. Client identity, product names, brand assets, proprietary prompts, business rules, model deployment names, credentials, and commercial metrics are excluded under NDA; outcomes here are described qualitatively.